Appendix –D (Effect of Foot Length)

Figure D1 shows the control effort required to move a foot/step length. The height of the obstacle climbed is zero (i.e., walking on a smooth surface). The left column (parts a, c, e) shows the control effort required for one step length (at a particular horizontal velocity). The right column (parts b, d, f) shows the control effort required to cover a total horizontal distance of 0.8 m (in multiple steps; however, horizontal velocity remains the same). Three distinct horizontal velocities have been considered, i.e., 0.26, 0.4, and 0.53 m/s. Similarly, Figure D2 shows the control effort required to achieve a given foot/step length. The height of the obstacle climbed is 5 mm. The left column shows the control effort required for one step length (at a particular horizontal velocity). The right column shows the control effort required to cover a horizontal distance of 0.8 m (in multiple steps; however, horizontal velocity remains the same). Three distinct horizontal velocities have been considered, i.e., 0.26, 0.4 and 0.53 m/s. Figure D3 shows the control effort required to move a step length with a horizontal velocity of 0.26 m/s and an obstacle height of 15 mm. The left-hand side represents the control energy required for one foot/step length (at a particular horizontal velocity). The right column shows the control effort required to cover a horizontal distance of 0.8 m (in multiple steps; however, horizontal velocity remains the same). In all three figures, the obstacle is of zero width and is located at the centre of the step length. It is clear from these figures that, for different horizontal velocities of the robot and different obstacle heights, there is an optimal foot/step length that minimizes the control energy requirement. It is observed that as the horizontal velocity increases, the optimal step length increases. It is also observed that as the height of the obstacle changes, the optimal step/foot length also changes.

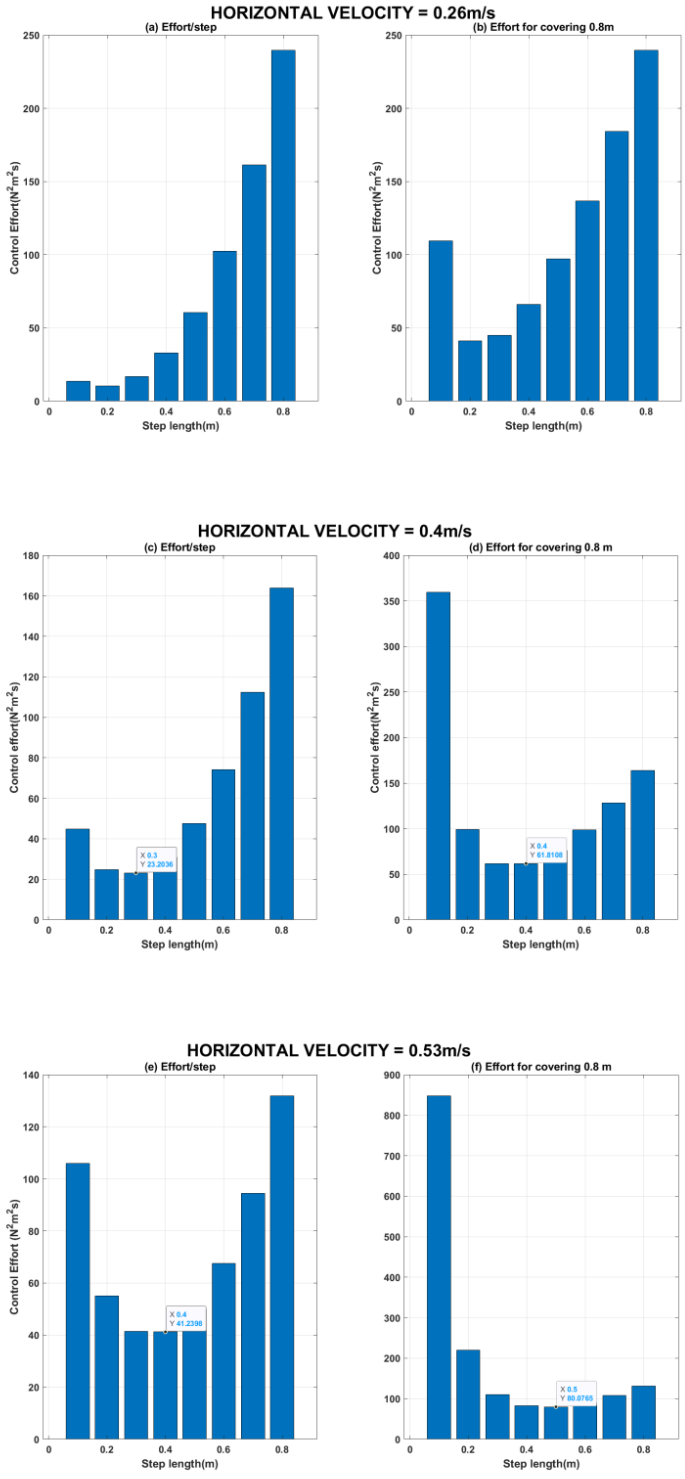


Figure D1: Optimal step length at different horizontal velocities (Height of obstacle =Zero)

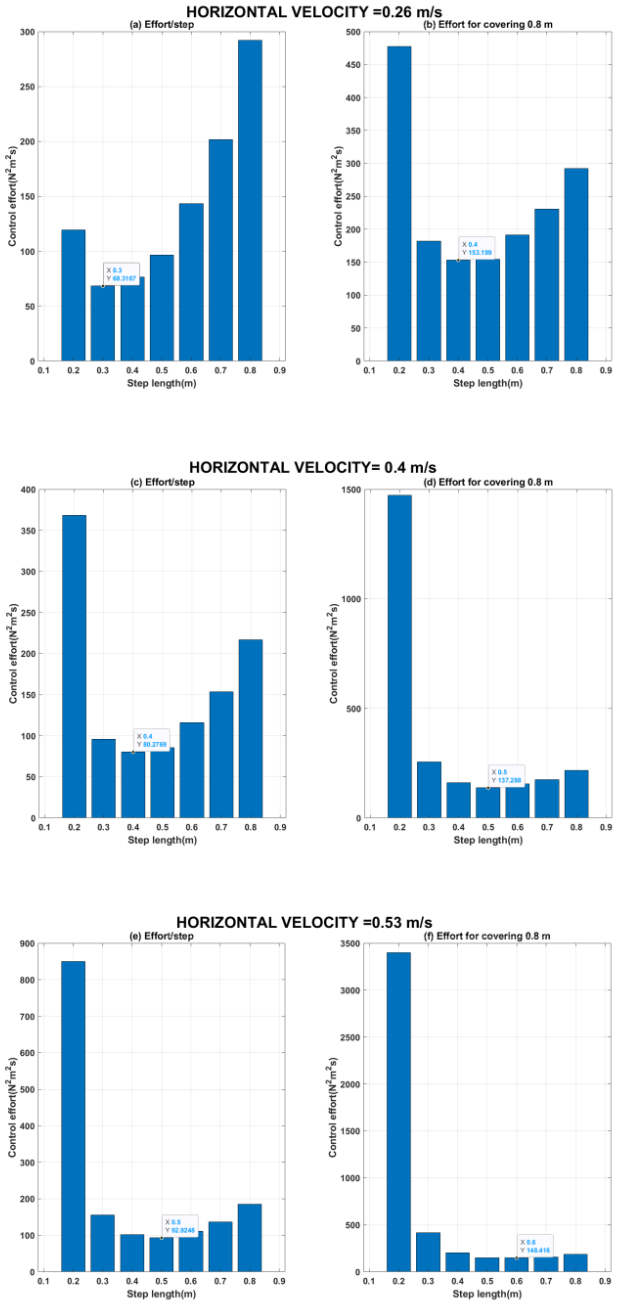


Figure D2: Optimal step length at different horizontal velocities (Height of obstacle =5 mm)

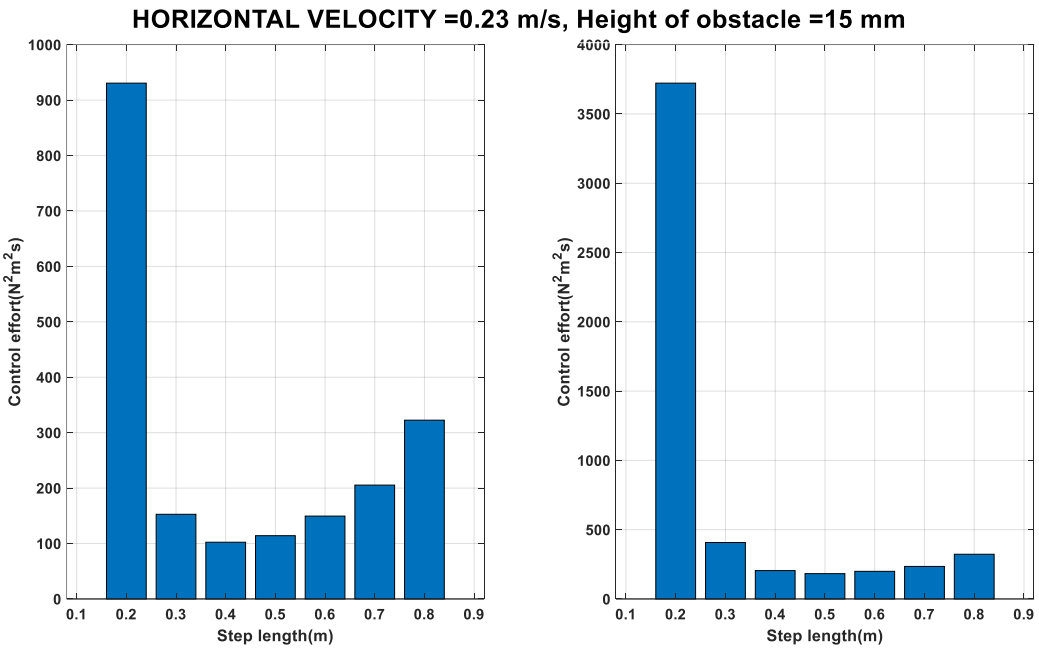


Figure D3: Optimal step length at Horizontal velocity=0.26 m/s and Height of obstacle =15 mm)